



Yug Sarda

Roll No.: 2023UCP1663

B.Tech in Computer Science and Engineering

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EDUCATION

•Malaviya National Institute of Technology, Jaipur

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 8.10/10

•Narayana Junior College

2021 – 2023

State Board (Intermediate)

Percentage: 97.8%

PERSONAL PROJECTS

•Autonomous Multi-Agent AI Research System

FastAPI, LangGraph, LlamaIndex, ChromaDB

- Engineered an autonomous multi-agent AI research platform using LangGraph-based orchestration for planning, retrieval, synthesis, critique, and self-reflective reasoning workflows.
- Built an advanced multimodal RAG pipeline using ChromaDB for semantic retrieval with hybrid retrieval, Multi-Query Retrieval, RAG Fusion, MMR diversification, and cross-encoder reranking for grounded research generation.
- Implemented citation-grounded answer generation with contextual compression, hallucination mitigation, confidence scoring, and iterative retry mechanisms for improved factual reliability.
- Developed short-term and long-term conversational memory using SQLite.
- Integrated LangSmith for agent workflow tracing and observability, along with Prometheus and Grafana for real-time system monitoring, latency analysis, performance metrics, and debugging.

•Healthcare Document Intelligence System

FastAPI, Neo4j, ChromaDB, Streamlit

- Designed a multimodal healthcare document intelligence system with separate processing pipelines for PDF and medical image inputs, leveraging Gemini Vision for OCR and visual understanding.
- Built a Hybrid Graph RAG architecture using Neo4j Knowledge Graphs, ChromaDB semantic retrieval, and patient memory to enable personalized, context-aware clinical question answering.
- Implemented asynchronous entity extraction pipelines using `asyncio.Queue`, enabling concurrent processing of clinical entities and improving document ingestion throughput.
- Developed fault-tolerant multi-agent workflows for document ingestion, entity extraction, graph construction, validation, and clinical reasoning with automatic recovery from pipeline failures.
- Engineered a clean FastAPI backend with complete separation of routing, business logic, and service layers, enabling modular, maintainable, and extensible AI workflows.

•Customer Churn Prediction System

Python, Pandas, Matplotlib

- Built a machine learning system to predict customer churn based on behavioral, demographic, and service-usage patterns.
- Performed extensive data preprocessing including missing-value imputation, categorical encoding, feature scaling, and outlier handling.
- Implemented and compared models such as Logistic Regression, Random Forest, and XGBoost.
- Evaluated the system using cross-validation, ROC-AUC, precision, recall, and confusion matrix for robust performance.

TECHNICAL SKILLS AND INTERESTS

Languages: C, C++, Python

Frameworks/Libraries: LangChain, LangGraph, FastAPI

Databases: MySQL

Developer Tools: Git, Postman, Google Colab, VS Code, Jupyter Notebook

Related Coursework: Machine Learning, Deep Learning, OOPS, DBMS, OS, DSA

Soft Skills: Team Work, Time Management, Communication, Adaptability

ACHIEVEMENTS

Secured 99.7 percentile and AIR rank 3402 in JEE Main and qualified for JEE Advanced

Selected in Amazon ML Summer School among 60000 applicants

Solved over 500+ DSA problems on LeetCode and GFG

LeetCode Knight Max Rating(1903)